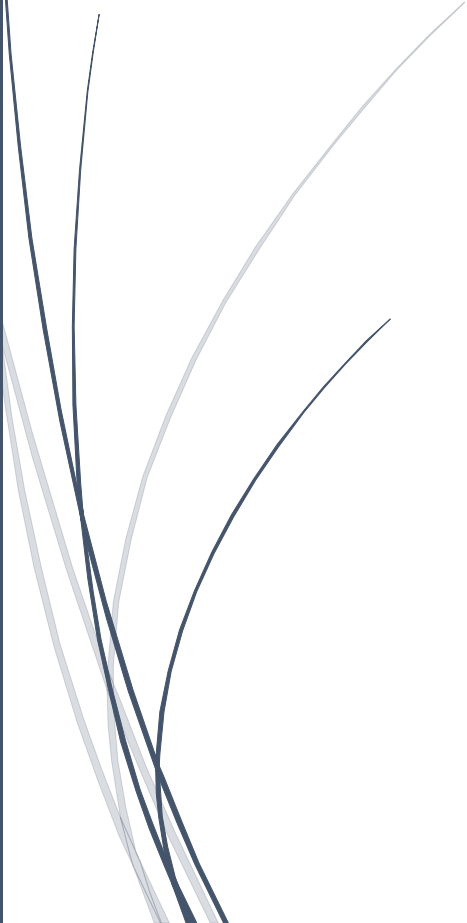


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29/05/2024

# Blockchain

## Cryptography in Blockchain

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## **Introduction:**

In a constantly evolving technological world, blockchain technology emerges as a revolutionary innovation built on a solid mathematical foundation.

The aim of this report is to closely examine the significant role of mathematics in blockchain concepts and how these mathematical principles underpin the security, authenticity, and trust at the core of this groundbreaking technology.

Starting by defining blockchain and emphasizing its essential relationship with mathematics, we will address key aspects of blockchain technology such as cryptographic addresses, smart contracts, consensus mechanisms, attacks, and countermeasures, use cases, and future perspectives.

Each step of this report demonstrates how mathematical concepts are integrated into the very structure of blockchain to ensure its integrity and robustness.

By exploring this fascinating intersection between mathematics and technology, we will uncover how cryptography, hashing, consensus algorithms, and other essential mathematical concepts form the foundation upon which the blockchain revolution rests.

# 1. Cryptographic Addresses:

Cryptographic addresses play an essential role in the Blockchain ecosystem, ensuring the unique identification of participants and the security of transactions. This section focuses on address generation and hash robustness, highlighting the importance of mathematical and cryptographic concepts in this crucial process.

## 1.1 Address Generation:

The generation of cryptographic addresses constitutes a crucial element of the blockchain ecosystem, ensuring unique participant identification and transaction security. This section delves into the detailed process of address generation, highlighting the importance of hash functions and digital signatures in guaranteeing the security and integrity of addresses within the blockchain.

### *Process of Generation:*

When a user wishes to engage in the blockchain network either as a participant or as a transaction recipient, a cryptographic address is assigned to them. This address is generated from a pair of keys: a public key and a private key. The private key is typically generated by selecting two large prime numbers and performing operations such as multiplication modulo a prime number, a fundamental concept of modular arithmetic in number theory. The resulting public key is derived from the private key using mathematical operations, ensuring a unique and secure identifier for the user within the blockchain network.

The basic technologies of cryptography include symmetric and asymmetric encryption. Asymmetric encryption, or public key encryption, solves the key distribution problem of symmetric encryption by using separate keys for encryption and decryption, called the public key and the private key. The private key is generated at random, while the public key is calculated irreversibly.

The advantages of asymmetric encryption include the separation of public and private keys for secure transmission over unsecured channels. However, it has disadvantages such as slower processing speed and reduced encryption power, requiring enhanced security. Elliptic curve cryptography, such as **secp256k1** used in blockchain, relies on the mathematical difficulty of the discrete logarithm of the elliptic curve to ensure security.

In the Bitcoin system, a key pair consisting of a private key and a public key is generated using public key encryption. The recipient's Bitcoin address is derived from their public key, ensuring secure transactions. The private key is a random number, while the public key is generated by encrypting the private key via an elliptic curve. A single-entry encrypted hash function is used to generate the bitcoin address from the public key. (Zhai, 2019)

### *Importance of Hash Functions:*

Hash functions play a pivotal role in the address generation process. A hash function takes an input (in this case, the public key) and generates a fixed-length output, known as a hash. This hash is then converted into an address in a human-readable form. Hash functions, based on mathematical principles from number theory and modular arithmetic, ensure data uniqueness, integrity, and confidentiality, which are essential for address security within the blockchain.

### *Digital Signatures:*

In addition to hash functions, digital signatures are also employed to ensure the security and integrity of addresses. When a transaction is initiated, it is signed with the sender's private key, thereby proving the authenticity of the transaction. This signature is verified by the blockchain network using the sender's public key, ensuring that the transaction originates from the authorized party.

By elucidating how hash functions bolster address security, we underscore the significance of these mathematical concepts in the reliability of the blockchain.

## 1.2 Hash Robustness:

Hash functions are vital for securing cryptographic addresses in the blockchain. By converting data into a fixed-length digital fingerprint, hash functions ensure data uniqueness, integrity, and confidentiality, thus strengthening blockchain security. These hash functions rely on various mathematical principles, including concepts from number theory, to provide robustness and security.

### **Securing Addresses:**

Hash functions are crucial for securing cryptographic addresses by making them unique and preventing data tampering. When a public key is processed through a hash function, it generates a unique digital fingerprint used as an address in the blockchain. This address is nearly impossible to reverse, making it extremely difficult to determine the original public key from the address. The security of these addresses relies on the difficulty of factoring large prime numbers and other mathematical operations derived from number theory.

### **Protection against Attacks:**

Hash functions also provide protection against attacks such as collisions, where two different inputs produce the same hash. The hash functions used in the blockchain are designed to minimize the risk of collisions, ensuring data integrity and transaction security. These functions rely on mathematical properties and algorithms derived from number theory to ensure the uniqueness and robustness of the generated hashes.

### **Data Confidentiality:**

In addition to ensuring data integrity, hash functions also preserve data confidentiality. When a transaction is broadcasted on the blockchain network, only the hashes of the data are publicly visible, preserving the confidentiality of sensitive details such as transaction amounts and the addresses of parties involved.

By illustrating how hash functions enhance address security, we emphasize the importance of these mathematical concepts in the reliability of the blockchain.

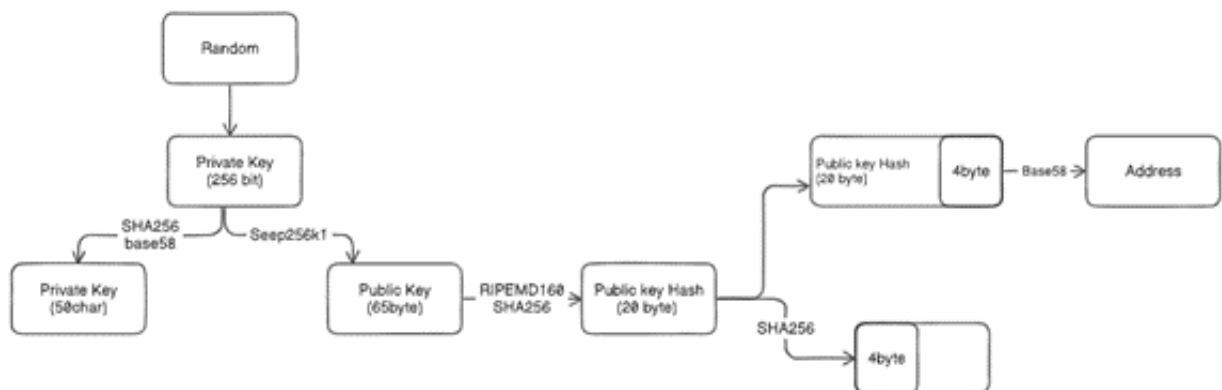


Figure 1: Address generation process of Bitcoin.

## 2. Smart Contracts:

Smart contracts are self-executing computer programs that execute predefined rules and conditions on the blockchain. This section highlights the definition and utility of smart contracts, as well as the cryptographic security that underpins their operation.

## 2.1 Definition and Utility

Smart contracts are computer protocols that facilitate, verify, and enforce the negotiation or execution of a contract or transaction. We will examine their formal definition and common use cases, emphasizing their role in automating processes and reducing transaction costs. Mathematically, smart contracts can be thought of as finite state machines, where each state represents a step in the contract, and each transition is conditional on specific events or logical conditions. The use of conditional logic (if-then-else) enables the contract to define the actions to be executed when certain conditions are met.

## 2.2 Cryptographic Security

Smart contract security relies on sophisticated cryptographic mechanisms to guarantee the integrity, confidentiality, and verifiability of transactions. We take a detailed look at the cryptographic algorithms and protocols used to guarantee smart contract security, highlighting the importance of hash functions, digital signatures, and asymmetric cryptography.

### ***Asymmetric cryptography:***

Each user has a pair of keys (a public key and a private key). The security of these keys is based on the mathematical difficulty of certain problems, such as the factorization of large prime numbers (RSA) or the discrete logarithm problem (elliptic curve cryptography, ECC). Transactions and interactions with smart contracts are digitally signed with the user's private key, guaranteeing data authenticity and integrity. Mathematically, this involves operations such as modular exponentiation and multiplication of elliptic curve points.

### ***Hash functions:***

Hash functions are used to guarantee data integrity. A hash function associates an input  $x$  with an output of fixed length  $h(x)$ . Any change to the input results in a significantly different output, known as the avalanche effect. Cryptographic hash functions, such as SHA-256, rely on properties such as image resistance (it is computationally impossible to find  $x$  given  $h(x)$ ), second image resistance and collision resistance (it is impossible to find two distinct inputs  $x$  and  $y$  such that  $h(x) = h(y)$ ).

### ***Digital signatures:***

Digital signatures can be used to verify the authenticity of a message or document. Using his or her private key, a user creates a signature that can be verified by anyone with the corresponding public key. From a mathematical point of view, a digital signature system involves key generation, signature, and verification algorithms. For example, in the RSA algorithm, the signature  $s$  of a message  $m$  is given by  $s = m^d \bmod n$ , where  $d$  is the private key and  $n$  is part of the public key. Verification consists of checking  $m = s^e \bmod n$ , where  $e$  is the public exponent.

### ***Game theory:***

This field is used to analyse, and design aligned incentives for network participants. Smart contracts are designed so that rational participants find it advantageous to follow the rules of the contract rather than cheat. Game theory studies strategies in competitive and cooperative situations. For example, in cooperative games, participants are rewarded for their cooperation, while in non-cooperative games, participants act to maximize their individual gains while adhering to the rules of the system. The concept of Nash equilibrium is often used to predict the outcome of these interactions.

### ***Temporal logic:***

Smart contracts can include temporal conditions, enabling actions to be triggered at specific times or after certain delays. Temporal logic involves reasoning about time and the temporal properties of events. For example, linear temporal logic (LTL) and computational tree logic (CTL) can be used to specify and reason about the order of events in smart contracts. If we take the example of Bitcoin and its timestamp server: the timestamp proves that the data must have existed at the time of the timestamp, of course, to obtain its digital fingerprint. Each timestamp includes the previous timestamp in its fingerprint, forming a chain, with each new timestamp reinforcing those preceding it. (NAKAMOTO, 2008)

## **3. Consensus Mechanisms:**

Consensus mechanisms are at the heart of any blockchain network, enabling participants to reach agreement on the current state of the distributed ledger. These mechanisms are based on sophisticated mathematical algorithms that guarantee the consistency and security of the network.

### **3.1 Consensus algorithms**

Consensus algorithms play a central role in the operation of blockchains by ensuring that all network participants reach agreement on the state of the blockchain, i.e. on the order of transactions and the data stored on it. Two of the most used consensus algorithms are Proof of Work and Proof of Stake.

#### ***Proof of Work:***

Proof of Work is the original consensus algorithm used by Bitcoin. In this system, miners must solve complex mathematical problems (called "puzzles") to validate and add new blocks to the blockchain. Solving these puzzles requires considerable computing power, ensuring that miners have invested significant resources (electricity and hardware) in the validation process. Once a miner has solved the puzzle, he or she broadcasts the newly validated block to the entire network, and if most nodes accept it as valid, it is added to the blockchain. Proof-of-work is appreciated for its robustness and security but is also criticized for its high energy consumption. (NAKAMOTO, 2008)

#### ***Proof of Stake:***

Unlike Proof of Work, Proof of Stake awards the right to validate and add new blocks to the blockchain based on the amount of cryptocurrency owned and staked by a participant. Instead of solving puzzles, validators (known as "stakeholders") must lock a certain amount of their cryptocurrency as "stake" to be eligible to validate transactions. Validators are randomly selected according to the amount of cryptocurrency they have staked. Proof-of-stake is valued for its lower energy consumption compared to proof-of-work but can be criticized for favouring participants with large amounts of cryptocurrency, potentially centralizing control of the network. (NAKAMOTO, 2008)

These consensus algorithms aim to solve the problem of double spending (i.e. ensuring that the same unit of cryptocurrency is not spent twice) and to ensure the security and integrity of the blockchain by enabling network participants to reach a consensus on the state of the ledger.



### 3.2 Role of Cryptography

Cryptography plays a crucial role in validating transactions and guaranteeing the integrity of the network. Transactions are digitally signed using private cryptographic keys, which guarantees their authenticity and origin. In addition, each block is linked to the previous one using a cryptographic hash function, creating a secure and immutable chain of blocks.

By using cryptography in innovative ways, blockchain networks can ensure the security and reliability of their transactions without the need for a trusted central authority. This decentralised approach guarantees the integrity of the network while preserving the confidentiality and anonymity of participants.

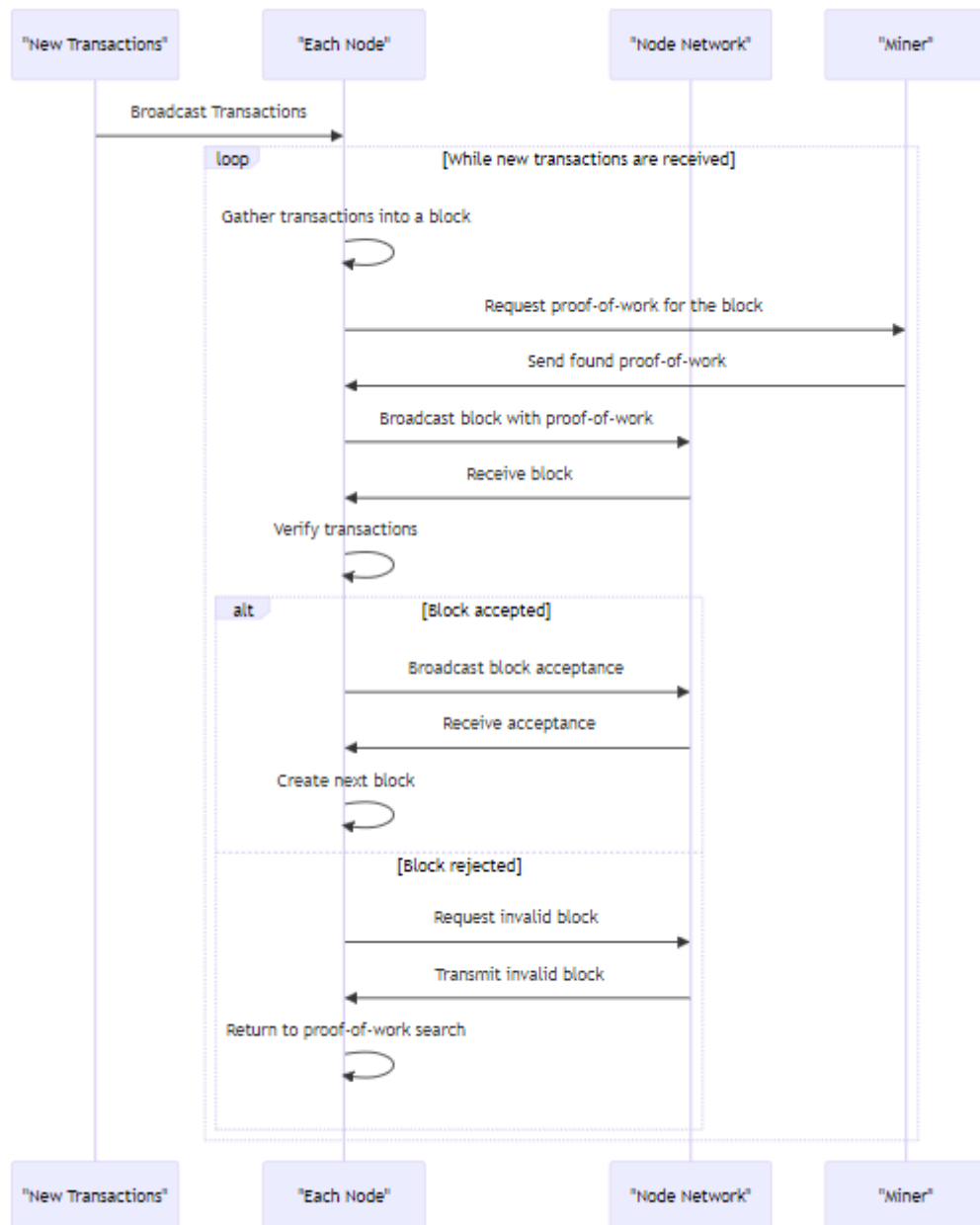


Figure 2: sequences diagram smart contract.

## 4. Attacks and Countermeasures:

In this section, we'll explore the various potential attacks that blockchain systems can face, highlighting the mathematical flaws. We will also look at countermeasures based on sound mathematical concepts to guard against these attacks (Yujuan Wen, 2021).

### 4.1 Attack analysis

Blockchain systems are the target of a variety of attacks due to their decentralized nature and the value of the assets they protect. Here are some of the most common attacks:

#### **51% attack**

Description:

This attack occurs when more than 50% of a network's computing power (or PoS ) is controlled by a single actor or collaborative group. This enables the attacker to manipulate the network, for example by cancelling transactions or preventing the confirmation of new transactions.

Mathematical concept:

Probability and Game Theory: The probability of a single player controlling more than 50% of resources is extremely low in a large, decentralized network. Game theory analyses the costs and benefits for participants of remaining honest or attacking the network.

#### **Double Spending Attack**

Description:

In a double spending attack, a user tries to spend the same unit of digital currency more than once. This can happen if an attacker succeeds in reorganizing the blockchain to include a different transaction in a block.

Mathematical concept:

Graph Theory: The blockchain's chain structure and the need for consensus ensure that confirmed transactions cannot be easily altered. Graph theory helps to understand the robustness of this structure against reorganization.

#### **Sybil attack**

Description:

In a Sybil attack, an attacker creates multiple fictitious identities to influence or take control of the network. This can be used to flood the network with fake transactions or to influence consensus processes.

Mathematical concept:

Network Theory: Network analysis can help identify anomalies in network structure, such as a suspicious concentration of nodes controlled by a single entity.

### 4.2 Countermeasures based on mathematical concepts.

To guard against attacks, various mathematical and cryptographic concepts are used to reinforce the security of blockchain systems.

#### **Countermeasures against the 51% attack**

Alternative Consensus Algorithms:

Proof of Stake (PoS) and Delegated Proof of Stake (DPoS): These consensus mechanisms reduce the risk of the 51% attack by making the attack economically unviable, as it would require the acquisition of a large amount of digital currency.

Advanced cryptography:

Verifiable Delay Functions (VDFs): Used to introduce verifiable delays, complicating the task of any actor trying to maliciously take control of the network.

### ***Countermeasures against the Double Spending Attack***

Transaction confirmation:

Number of confirmations: impose a minimum number of confirmations before a transaction is considered valid and irreversible. The higher the number of confirmations, the more difficult it becomes for an attacker to reorganize the blockchain.

Consensus algorithms:

Probabilistic Consensus Algorithms: These algorithms reduce the probability of success of a double spending attack by exponentially increasing the difficulty of modifying older blocks.

### ***Countermeasures against Sybil's attack***

Identity systems:

Proof of Identity (PoID): Use identity verification systems to guarantee that each network participant is a unique entity.

Network Security Protocols:

Anomaly Detection Algorithms: Use machine learning and network analysis algorithms to detect and block Sybil attacks in real time.

By analysing potential attacks and proposing sound mathematical countermeasures, this section shows how mathematical concepts are key to ensuring the security and resilience of blockchain systems.

## **5. Use Cases:**

Blockchain has applications in many fields, thanks to its unique properties of security, transparency, and decentralization. Here are a few concrete examples:

**Supply Chain Management:** Used to track products from origin to destination, guaranteeing data integrity through hash functions and digital signatures. For example, IBM Food Trust uses blockchain to improve food traceability and reduce food fraud.

**Electronic voting:** Enables secure and transparent voting systems, reducing the risk of electoral fraud. Zero-knowledge proofs and asymmetric cryptography guarantee that each vote is unique and anonymous.

**Digital Identity:** Provides a secure, immutable digital identity, giving users complete control over their personal information. Civic, for example, uses blockchain to securely store and share identity information.

**Smart Contracts:** Ethereum enables the deployment of smart contracts, self-executing programs that guarantee the correct and secure execution of transactions through digital signatures and hash functions.

## **Conclusion:**

In conclusion, this report has highlighted the crucial importance of mathematics in the field of Blockchain. Through a thorough exploration of cryptographic addresses, smart contracts, consensus mechanisms, attacks, and countermeasures, as well as various use cases, we have demonstrated how mathematical foundations are at the heart of the blockchain revolution.

By connecting mathematical concepts of cryptography, hashing, consensus algorithms, and other key principles to every aspect of the Blockchain, we have emphasized how these foundations ensure security, transparency, and trust within this decentralized network. Mathematics thus becomes the cornerstone upon which the reliability and integrity of the Blockchain rest, paving the way for new possibilities and innovations in an ever-evolving digital world.

Looking ahead, mathematical advancements will continue to shape and strengthen the Blockchain, providing more robust, secure, and efficient solutions for a multitude of applications. By fully embracing the marriage between mathematics and technology, we witness a profound transformation that redefines how we conceive, exchange, and interact with digital data and transactions.

Thus, this report eloquently underscores that mathematics is not just a tool, but the very foundation upon which the promising future of the Blockchain rests. By uniting these two constantly evolving domains, we are ushering in an era of enhanced trust, continuous innovation, and endless possibilities in the world of blockchain technology.

## Figures

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## Reference's

NAKAMOTO, S. (2008). *Bitcoin: A Peer-to-Peer Electronic Cash System*.

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